Synthetic Instance Configuration 800–1000

Generated Files

Instance file	$\|S\|$	$\|P\|$	$\|B\|$	$\|E\|$	$\|T\|$	$\|I\|$	Seed
problem_instance_ext_800_0.txt	800	960	117	5	10	117	1250
problem_instance_ext_800_1.txt	800	960	129	6	15	129	1251
problem_instance_ext_800_2.txt	800	960	126	6	8	126	1252
problem_instance_ext_800_3.txt	800	960	131	6	18	131	1253
problem_instance_ext_1000_0.txt	1000	1200	170	6	7	170	1254
problem_instance_ext_1000_1.txt	1000	1200	135	2	4	136	1255
problem_instance_ext_1000_2.txt	1000	1200	129	6	6	129	1256
problem_instance_ext_1000_3.txt	1000	1200	130	2	5	130	1257

$|S|$: students; $|P|$: placements; $|B|$: companies; $|E|$: educational centres; $|T|$: professors; $|I|$: dependency groups; Seed: seed used to generate the instance.

Generator Arguments

Argument	Description	Value
result_path	Output folder where the generated files are written	output
start_size	First number of students considered	800
end_size	Last number of students considered	1000
number_instances	Number of instances generated for each student size	4
surplus	Extra placements percentage with respect to students	20
slack_ratio	Extra capacity margin for company and professor quotas	0.2
placement_dep_prob	Percentage used to create placement dependency groups	30
number_competences	Number of base competences per profile	6
most_important_competences	Number of primary competences required by a placement	2
seed	Base seed used to derive the seed of each generated instance	1250

Internal Parameters

Parameter	Description	Value
student_size_step	Fixed step between generated student sizes	200
number_profiles	Number of academic profile blocks	8
p_excellent_student	Probability of generating an excellent student	0.05
p_notable_student	Probability of generating a notable student	0.25
car_probability	Probability that a student has access to a car	0.65
student_distance_weight_range	Range used for the student distance weight	$[0.0, 0.5]$
n_centers_min	Minimum number of educational centres	2
n_centers_max	Maximum number of educational centres	6
prof_capacity_min	Minimum professor supervision capacity	5
prof_capacity_max	Maximum professor supervision capacity	25
ORS_AVAILABLE	Whether ORS routing support is available in the script	False
use_ors	Whether ORS is actually used for this generation	False
ors_base_url	ORS endpoint used when routing is enabled	http://localhost:8080/ors
synthetic_duration_mu	Mean of the synthetic duration distribution	1500
synthetic_duration_sigma	Standard deviation of the synthetic duration distribution	250

Randomness

Field	Description	Value
Base seed	Initial seed provided to the generator	1250
Seed policy	Rule used to derive instance seeds from the base seed	instance_seed = base_seed + global_instance_idx
Python random seeded	Whether the Python random generator is seeded per instance	Yes
NumPy random seeded	Whether the NumPy random generator is seeded per instance	Yes